Bogdan Dukic

Machine Learning | Computer Vision | Computer Graphics | Research Engineering
Palaiseau, France | bogdandjukic25@gmail.com | +33 7 46 41 82 15
bgdj11.github.io | GitHub: bgdj11 | LinkedIn: bogdandjukic25

Portfolio

EDUCATION

École Polytechnique, Institut Polytechnique de Paris Sep 2026 - 2028 (expected)
MScT in Visual and Creative Artificial Intelligence
Coursework: Foundations of Machine Learning, Computer Vision, Shape Analysis, Computer Animation.
Upcoming: Advanced Deep Learning, Multimodal Generative AI, Image Synthesis.
Scholarship: École Polytechnique Foundation Excellence Scholarship (2026-2028).

Faculty of Technical Sciences, University of Novi Sad Oct 2021 - Jul 2026
BSc in Computing and Control Engineering | Intelligent Systems | GPA 9.55/10
Coursework: Computational Intelligence, Knowledge-Based Systems, Soft Computing, Optimization, Data Structures.
Scholarships: Young Talents of Serbia (Dositeja, 2025), Don Walker - Magna (2022-2025), Ministry of Education (2022-2024).

SELECTED PROJECTS & RESEARCH

Neuro-Symbolic Visual Scene Understanding Bachelor's thesis | 2025 - 2026
Python, PyTorch | Full thesis (SR) | Research summary (EN)
- Designed a modular VQA system connecting neural scene perception with a 27-operator typed DSL, program validation and a deterministic executor exposing intermediate reasoning steps.
- Fine-tuned T5-small for question-to-program translation and adapted it to human-written questions using REINFORCE, execution rewards and KL regularization without target-domain program annotations.
- Improved CLEVR-Humans end-to-end accuracy from 32.7% to 45.6% on 7,202 questions and reduced execution failures from 25.5% to 0.6% in the annotated-scene evaluation track.
- Evaluated CLEVR, CoGenT and CLEVR-Humans to separate perception and reasoning errors. Achieved 99.9% CLEVR accuracy with annotated scene graphs and 79.0% end-to-end.

Federated Actor Learning 2026
- Built an asynchronous actor runtime with local/TCP messaging, supervision and SQLite persistence, supporting coordinated FedAvg and decentralized CRDT-based gossip training.
- Evaluated both workflows on non-IID EuroSAT partitions using ResNet18 features and tested model exchange, aggregation, state merging and checkpoint recovery.

Photometric Stereo for 3D Surface Reconstruction 2026 - Present
- Developing a photometric stereo pipeline for surface normal and height-field reconstruction, with least-squares and robust IRLS baselines on Blender-rendered scenes.
- Exploring illumination models, adaptive light selection and implicit neural representations.

PROFESSIONAL & RESEARCH EXPERIENCE

AI Engineer | Comtrade System Integration Jan - Aug 2026 | Belgrade
- Built document ingestion and RAG pipelines with Docling/OCR, semantic chunking, hybrid retrieval and reranking for Azure and on-premises environments.
- Benchmarked Serbian and multilingual LLMs and embeddings, adding grounding, confidence routing and fallback mechanisms.
- Developed a no-code voice-agent prototype integrating modular LLM, speech-to-text (STT) and text-to-speech (TTS) services with configurable conversational guardrails.
- Developed call analytics and evaluated Serbian speech solutions. Contributed to a React visual agent builder and an agenda-aware Microsoft Teams assistant.

Data Science Intern | Daze Solutions Limited Dec 2024 | Remote
- Built a Python pipeline for real-estate data extraction, cleaning, normalization and feature engineering.
- Applied K-means clustering and visualized property groups to identify shared characteristics.

Research Intern | Petnica Science Center Apr 2019 - Sep 2020 | Petnica
- Conducted biology and chemistry laboratory experiments, including hypothesis formulation, data interpretation, scientific writing and presentation of findings.

TECHNICAL SKILLS & LANGUAGES

Programming: Python, C++, Java, SQL.
ML & vision: PyTorch, Transformers, OpenCV, scikit-learn, NumPy, Pandas, FAISS.
Development: Spring Boot, FastAPI, React, Angular.
Tools & infrastructure: Git, Linux, Docker, Microsoft Azure, PostgreSQL.
Mathematics: Linear algebra, probability and statistics, optimization, numerical methods.
Languages: Serbian (native), English (C1, TOEFL iBT 100/120), French (actively learning).